Fundamentals of Machine Learning (SEng4091)



Compiled By: Alemisa E.

Course Learning Outcome(CLOs)

❖ Upon successful completion, this course is designed to:

- Understand the mathematical foundations underlying ML algorithms and its major learning paradigms.
- Implement appropriate supervised, unsupervised learning and Reinforcement learning algorithms to solve real-world problems.
- Evaluate and compare machine learning models using appropriate validation methods and performance metrics.
- Explain, design, implement, train, and evaluate deep learning models for solving real-world machine learning problems.
- Apply data preprocessing and feature engineering techniques to prepare datasets for machine learning.
- Develop and optimize machine learning models through algorithm selection, parameter tuning, and model evaluation.
- Analyze ethical, fairness, privacy, bias, and generalization issues in machine learning applications.

Preliminaries Required:

- Here are some of the basic prerequisites that will help get started with ML.
  - *Programming (python, Matlab, R etc…)*
  - *Probability and Statistics*
  - *Mathematics (Linear Algebra, Matrix, Vector, differential calculus, and integral calculus)*

Textbook: ***Bishop, C. (2006), Pattern Recognition and Machine Learning, Berlin: Springer-Verlag***



Chapter-One

Introduction to Machine Learning

Outline

- Introduction
- Why Machine Learning (ML)?
- What is Machine learning?
- ML relationships to other fields
  - ✓ AI vs ML vs DL
- Essential math and statistics for machine learning
- Applications of machine learning
- Types of machine learning techniques



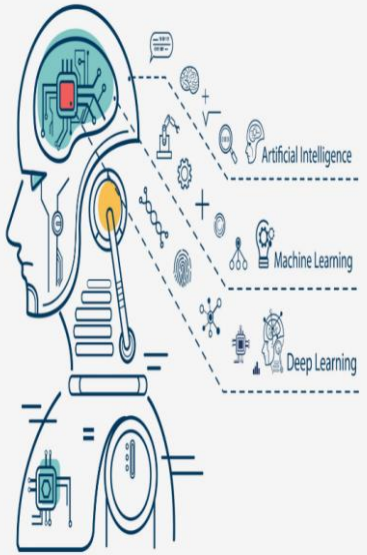
Introduction to ML

❖*Machine Learning is one of the most exciting technologies that one would have ever come across.*

❖ML is a branch of AI that concerns the construction and study of systems that can ***learn from data*.**

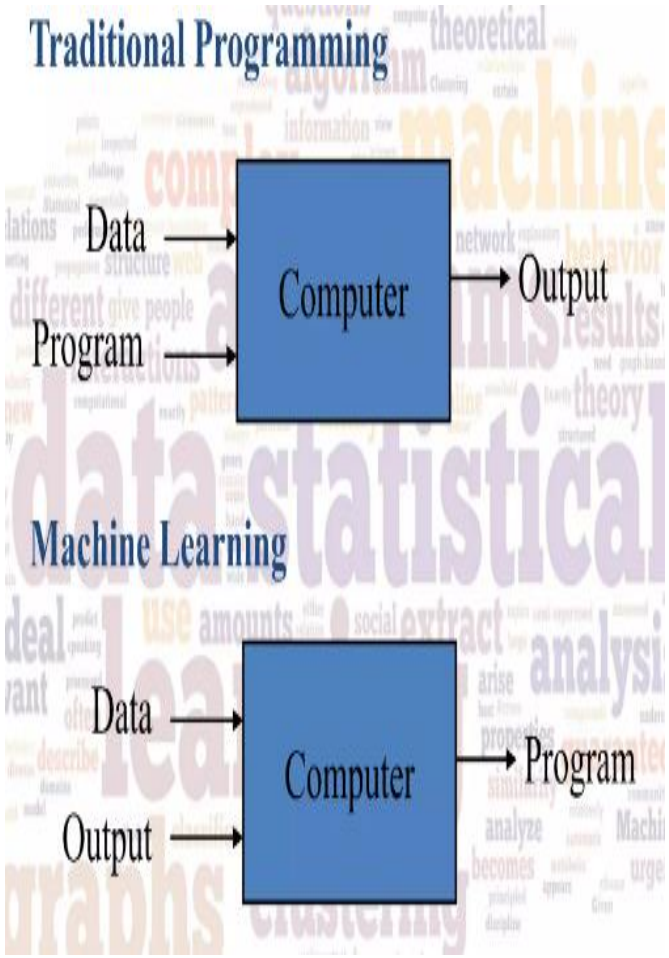
❖As intelligence requires ***knowledge,*** it is necessary for computers to ***acquire knowledge*.**

❖ML refers to a system capable of the autonomous acquisition and integration of knowledge.



❖***Machine learning*** *is actively being used today, perhaps in many more places than one would expect.*

Why Machine Learning (ML)?



❖**Traditional approach.**

- ❑ If the problem is not trivial, your program will likely become a long list of complex rules pretty hard to maintain.
- ❑It is *very hard* to write programs that solve *a problem*.
- ❑We *don't know what program to write* because we *don't know how* its done in our brain.
- ❑Even if we had a *good idea about how to do it*, the program might be *terribly complicated*.

❖**Machine Learning approach.**

- ❑The program is much shorter, easier to maintain, and most likely more accurate.

Why Machine Learning (ML)? …

Here is the few reason,

Why ML?







Why Machine Learning (ML)? …

❖ *You have probably use a learning algorithm dozens of times a day without knowing it.*

❖ *For example:*

- *Every time you use a web search engine like Google or Bing to search the internet, one of the reasons that works so well is because a learning algorithm, one implemented by Google or Microsoft, has learned how to rank web pages.*
- *Every time you use* **Facebook** *or Apple's photo typing application and it recognizes your friends' photos, that's also machine learning.*
- Every time you read your email and your spam filter saves you from having to wade through tons of spam email, that's also a learning algorithm.

Why is ML so prevalent Today?

❖Machine Learning:

❖*Grew out of the field of artificial intelligence.*

- ❑We wanted to build intelligent machines and it turns out that there are a few basic things that we could program a machine to do such as ***how to find the shortest path from A to B.***
- ❑But for the most part we just did not know how to write AI programs to do the more interesting things such as ***web search*** *or* ***photo tagging*** *or* ***email anti-spam.***
- ❑*There was a realization that the only way to do these things was to have a* ***machine learn to do it by itself.***

❖ New capability for computers and today it touches many segments of industry and basic science.

Some other Examples of ML

❖**Database mining.**

- Large datasets from growth of automation/web.
- for example, tons of Silicon Valley companies are today collecting web click data, also called clickstream data, and are trying to use ML algorithms to mine this data to understand the users better and to serve the users better

❖ **Medical records:**

- if we can turn medical records into medical knowledge, then we can start to understand the disease better.

❖**Computational biology:**

- With automation again, biologists are collecting lots of data about gene sequences, DNA sequences, and so on, and machines running algorithms are giving us a much better understanding of the human genome, and what it means to be human.

❖**Engineering:**

- In all fields of engineering, we have larger and larger, data sets that we're trying to understand using learning algorithms.

Some other Examples of ML...

❖**Applications can't program by hand:**

- ❑*For example, is somebody worked on autonomous helicopters for many years. He/she just did not know how to write a computer program to make this helicopter fly by itself. The only thing that worked was having a computer learn by itself how to fly this helicopter.*

❖***Most of natural language processing(NLP), computer vision etc.***

- ❑Most of NLP and most of computer vision today is applied machine learning.
- ❑ Learning algorithms are also widely used for self-customizing programs.
- ❑ Every time you go to Amazon or Netflix or iTunes Genius, and it recommends the movies or products and music to you, that's a learning algorithm.

Formal Definition of ML

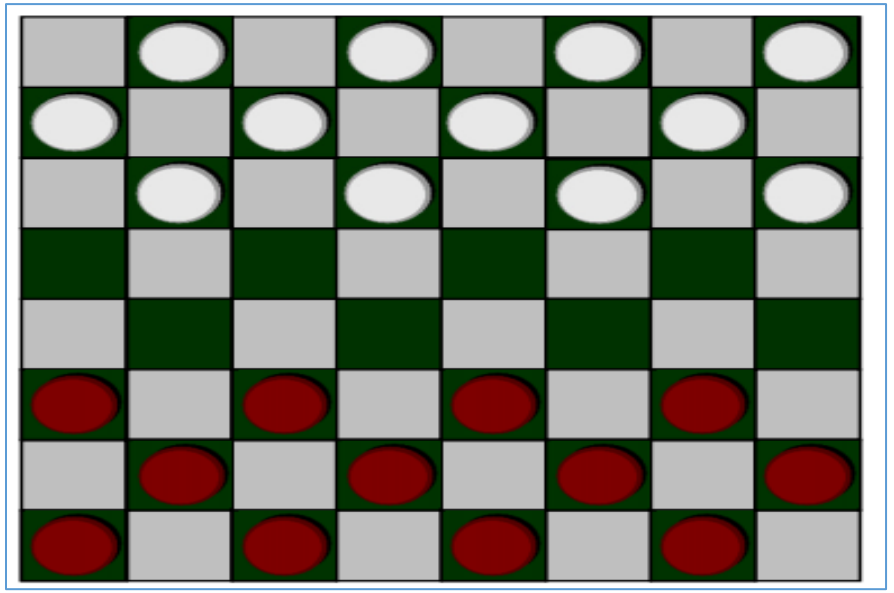
What is Machine Learning?

❖This section try to define: What it is ML? When we want to use ML? What isn't ML?
❖But let me show you a couple of examples of the ways that people have tried to define it.

Checkers Playing Program

❖**Arthur Samuel (1959).**

He defined *machine learning* as the field of study that gives computers the ability to learn without being explicitly programmed.



Definition of ML by Tom Mitchell: Example

❖ **Tom Mitchell (1998) Well-posed Learning Problem:**

" A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E."

Example #1

- **Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam.**
- **What is the task T in this setting?**

A. Classifying emails as spam or not spam. → T
B. Watching you label emails as spam or not spam. →E
C. The number (or fraction) of emails correctly classified as spam/not spam. → P
D. None of the above—this is not a machine learning problem

Example #2: Handwriting recognition learning problem

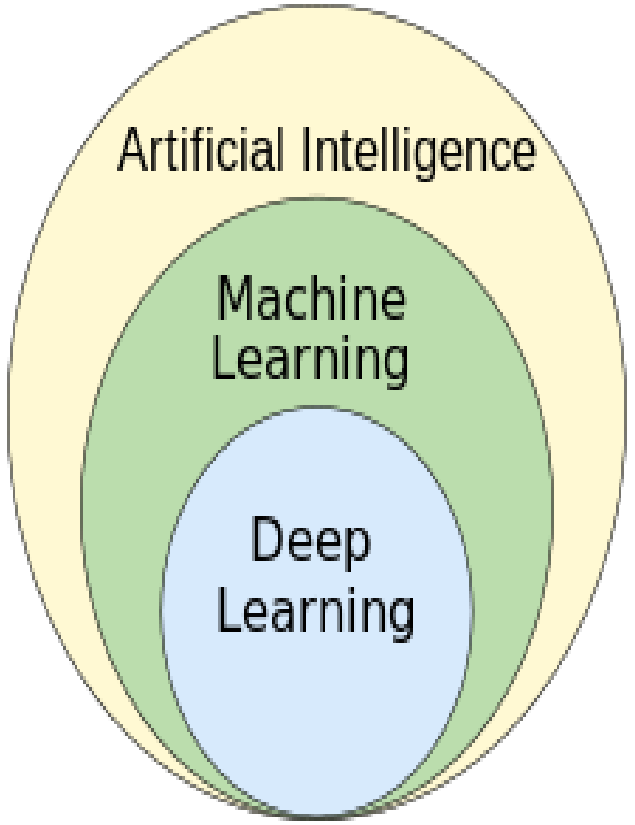
- **Task T :** Recognizing and classifying handwritten words within images
- **Performance P :** Percent of words correctly classified
- **Training experience E :** A dataset of handwritten words with given classifications

Example #3: A robot driving learning problem

- **Task T :** Driving on highways using vision sensors
- **Performance P :** Average distance traveled before an error
- **Training experience E :** A sequence of images and steering commands recorded while observing a human drive

Relationships to other fields

Concept	Definition
Artificial intelligence:	The aims is to create intelligent machines that can think and function like humans.
Machine learning:	A subfield of AI that focuses on developing algorithms and models that can learn from data rather than being explicitly programmed.
Deep learning:	A subfield of ML that uses multi-layered artificial neural networks to learn complex patterns in data.



Machine Learning Vs AI

- ***Machine learning*** **:** is more of *induction* (from specific to *generalization*)
  - Abebe is mortal, Almaz is mortal
  - Kebede is mortal,
  - Then, all humans are mortal
- ***Artificial Intelligence:*** more of *deduction* (from general to specific – *reasoning*)
  - A implies B
  - Having "A", you can infer "B"

• *Data* is the key in ML:

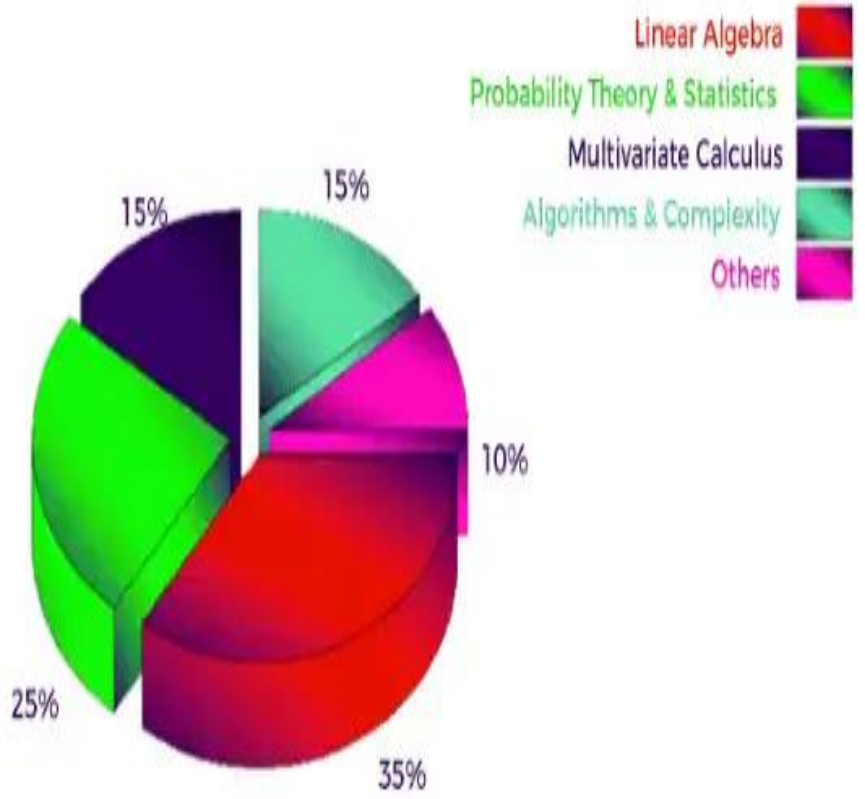
❖ The difference b/n solving a problem as *AI* or *ML* person is:

- As a *AI person*: the *algorithm* is central
- As a *ML person*: the *data* is central (get the priority)

Essential Math and Statistics for ML

- Machine Learning is currently one of the most popular technologies.
- But, what is the secret behind ***Machine Learning?***

- *The response is* ***Mathematics.***



Importance of Maths Topics Needed For Machine Learning

Essential math and Statistics for ML …

❖ Linear Algebra for Ml

❖**It is capable of the following:**

- ❑The study of vectors and some of the principles for manipulating ML
- ❑Used to develop ML algorithm
- ❑Data optimization is made easier via linear algebra.
- ❑It is utilized in support vector machine(SVM), classification, covariance matrices, regularization, loss functions, and regularization.
- ❑ It is utilized in ML linear regression.

❖ Calculus for Ml

❖Calculus is a key component in Ml.

❖ML uses the concepts of calculus to formulate the functions that are used to train algorithms.

❖ML models are trained with datasets having multiple feature variables.

❖Hence, getting familiar with multivariable calculus is important for building a suitable model.

- Some crucial areas of multivariate calculus: *Partial Derivatives*, *Vector-Values Functions*, *Directional Gradient*,

Essential math and Statistics for ML...

❖ Statistics in MI

❖**Statistics** is the main part of mathematics for ML.

❖Following are some basic statistical ideas that are required for machine learning:

✓*Combinatory, Axioms, Bayes' Theorem, Variance and Expectation, Random Variables, Conditional, and Joint Distributions.*

❖ Discrete Mathematics in ML

❖involves the use of discrete (non-continuous) numbers, most frequently integers.

❖Numerous domains, including algorithms, programming languages, cryptography, software development, ML and artificial intelligence applications etc., use DM extensively.

❖ For instance, a neural network can have 56 nodes and an integer number of nodes and connections.

❖ Graph algorithms and graph structure are two significant discrete mathematics topics for ML.

Applications of Machine Learning

- Today, companies are using ML to *improve business decisions, increase productivity, detect disease, forecast weather, and do many more things.*
- **Some of the most common ML Applications:**
  - Image Recognition:
  - Speech Recognition:
  - Recommender Systems:
  - Self Driving Cars
  - Medical Diagnosis
  - Stock Market Trading

Some of the most Common ML Applications:

❖Image Recognition:

- ✓The task which started from classification between cats and dog images has now evolved up to the level of Face Recognition and real-world use cases based on that like **employee attendance tracking.**
- ✓image recognition has helped revolutionized the healthcare industry by employing smart systems in disease recognition.

❖Recommender Systems:

- ✓ An example of these services is very common for example **YouTube.**
- ✓ It recommends new videos and content based on the user's past search patterns.
- ✓**Netflix recommends** movies and series based on the interest provided by users when someone creates an account for the very first time.

Some of the most common ML Applications...

❖Speech Recognition:

- ✓Based smart systems like **Alexa** and **Siri** have certainly come across and used to communicate with them.
- ✓ Converting voice instructions into text.

❑Self Driving Cars

- ✓The most common example of this use case is that of the Tesla cars which are well-tested and proven for autonomous driving.

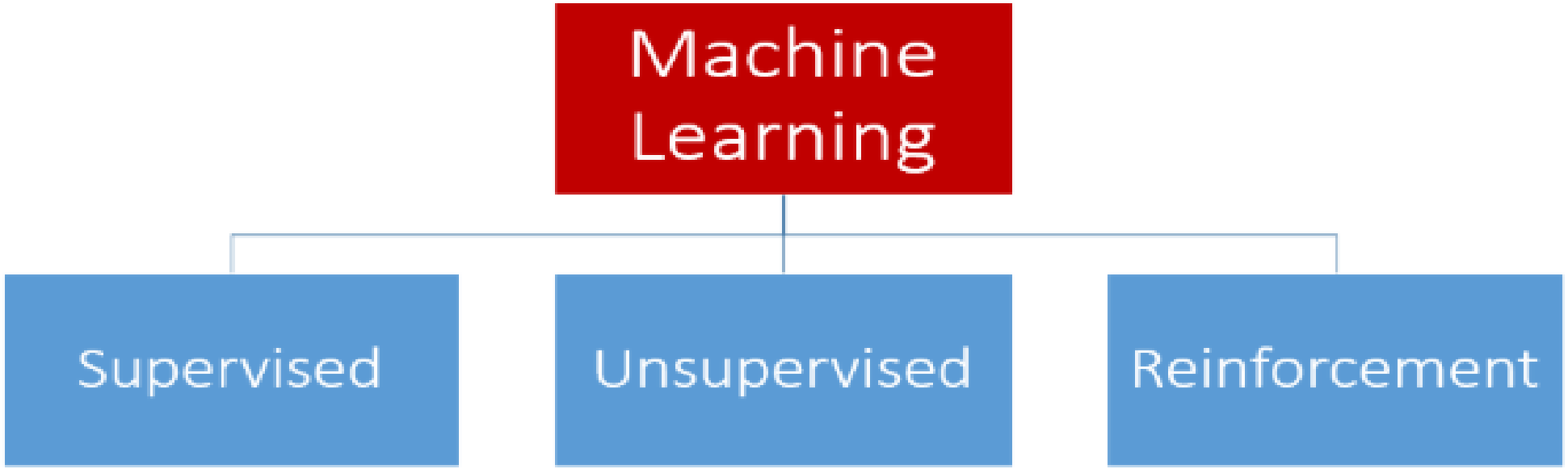
❑Medical Diagnosis

- ✓Many more health-related tasks are performed by machine learning models
- ✓For example cancer Classification,

❑Stock Market Trading:

- ✓to create intelligent systems that can predict future price trends and market value as well.

Types of Machine Learning techniques



Supervised Learning
- X,y (pre-classified training examples)
- Given an observation x, what is the best label for y?

Unsupervised learning
- X
- Given a set of x's, cluster or summarize them

Supervised learning

- **Supervised learning** is a class of problems that uses a model to learn the mapping between the ***input*** and ***target variables***.
- Applications consisting of the training data describing the various input variables and the target variable are known as ***supervised learning tasks.***
- Let the set of input ***variable be (x)*** and the **target variable be (y)**.
- A supervised learning algorithm tries to learn a hypothetical function which is a mapping given by the expression **y=f(x),** which is a function of x.

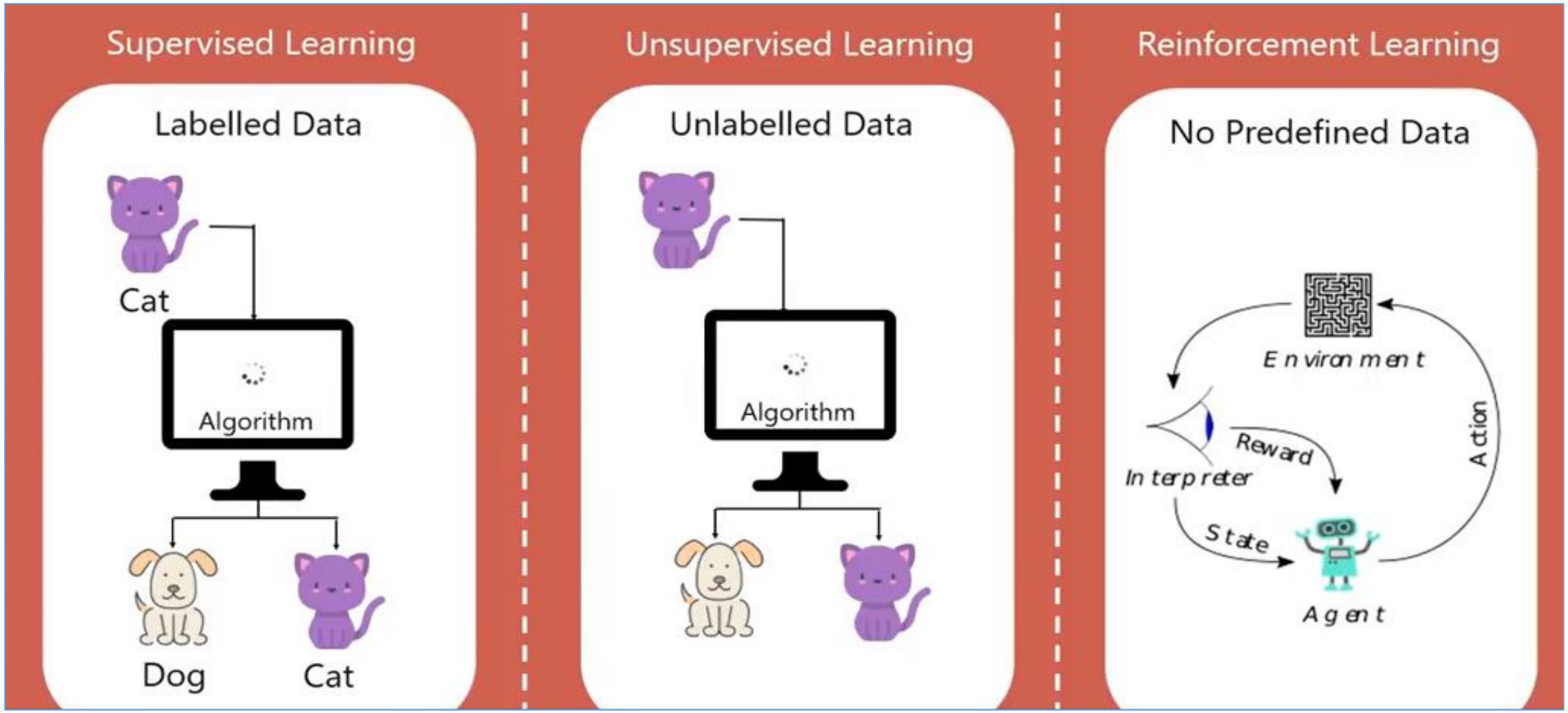
Unsupervised learning

- In an unsupervised learning problem the model tries to learn by itself and recognize patterns and extract the relationships among the data.
- As in case of a supervised learning there is no supervisor or a teacher to drive the model.
- Unsupervised learning operates only on the input variables.
- There are no target variables to guide the learning process.
- The goal here is to interpret the underlying patterns in the data in order to obtain more proficiency over the underlying data.

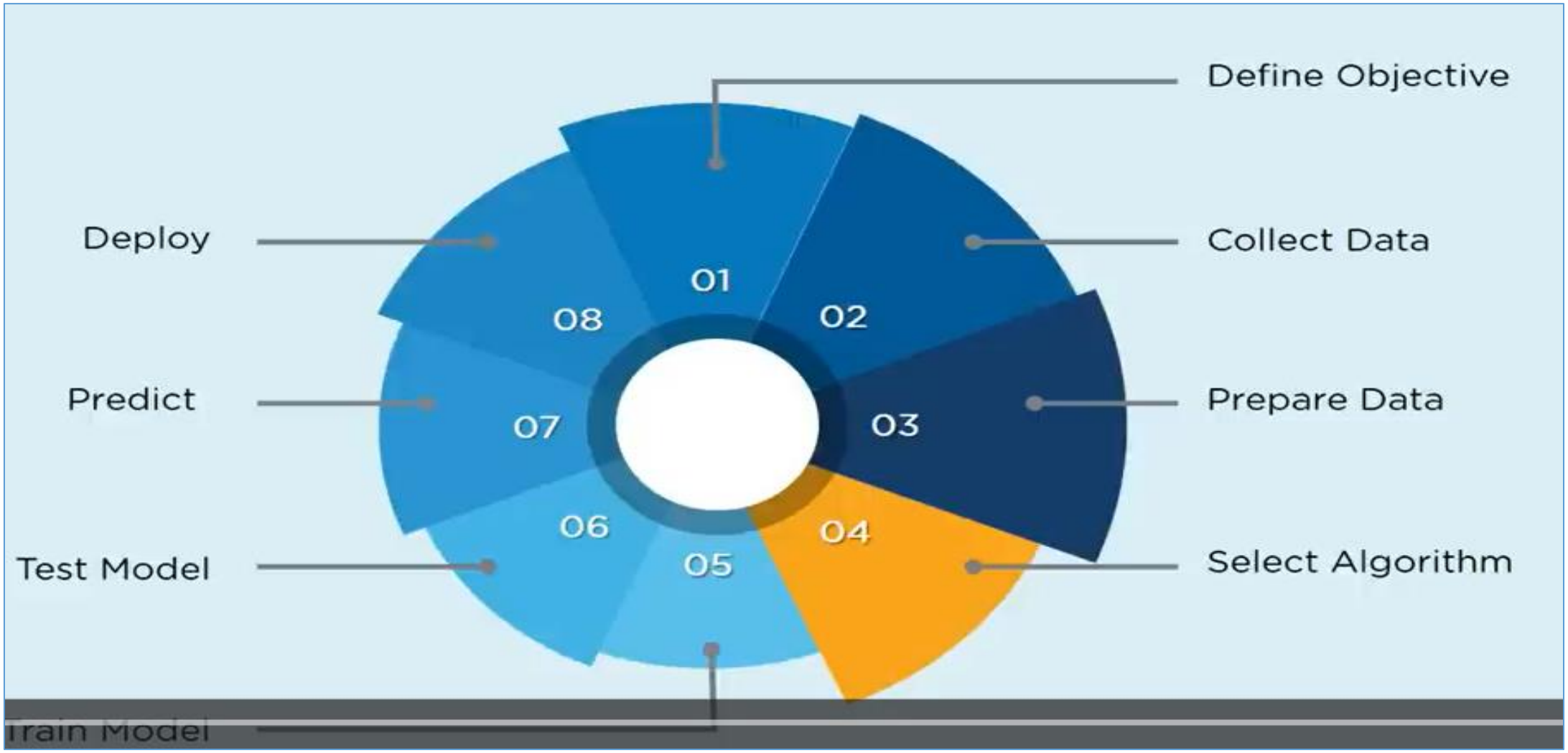
Reinforcement learning

- **Reinforcement learning** is type a of problem where there is an agent and the agent is operating in an environment based on the feedback or reward given to the agent by the environment in which it is operating.
- The rewards could be either positive or negative.
- The agent then proceeds in the environment based on the rewards gained.
- The reinforcement agent determines the steps to perform a particular task.
- There is no fixed training dataset here and the machine learns on its own.

Types of ML techniques: Example



How to get started with ML?



Question & Answer



Thank You !!!